Career 247

CONNECTTEL

Advanced Churn Prediction Strategy

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Project Overview & Business Context

Defining the challenge of reactive vs. proactive retention.

THE CONNECTTEL MANDATE

ConnectTel faces a significant challenge with increasing churn in a highly competitive telecom landscape.

- Identify high-value customers at risk.
- Move from reactive to proactive retention.
- Quantify and resolve friction points.

SOLVING FOR COMPLEXITY

Silent Churn

Identifying high-value users who stop engaging digitally but haven't yet initiated a formal complaint.

Financial Friction

Detecting "Bill Shock" trends where overage charges relative to tenure create sudden dissatisfaction.

THE ANALYTICAL JOURNEY

Q1

EDA & Data Ingestion

Q2

Feature Engineering

Q3

Ensemble Development

Q4

Strategic Optimization

DATA ENGINEERING EXCELLENCE

Velocity

Calculated change in monthly spending relative to customer lifetime history.

Friction

Combined metric of network issues and complaint frequency to score pain.

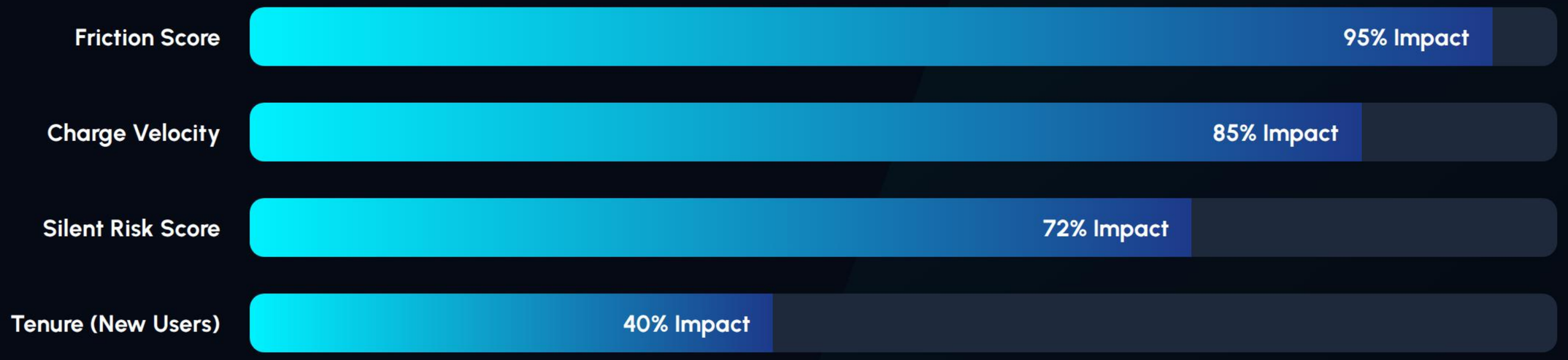
Silent Risk

Proprietary score mapping usage volume against lack of app engagement.

MODELING EVOLUTION

Architecture	Baseline	Optimized	Recall
Logistic Regression	58%	61%	42%
Random Forest	62%	68%	55%
XGBoost Champion	64%	**74%**	**82%**

TOP DRIVERS OF CHURN



Friction and price velocity represent the primary catalysts for customer exit.

MODEL CONFIDENCE

88%

ROC-AUC Metric

Our ensemble demonstrates an 88% probability of correctly ranking a random churner, allowing for surgical precision in retention marketing.

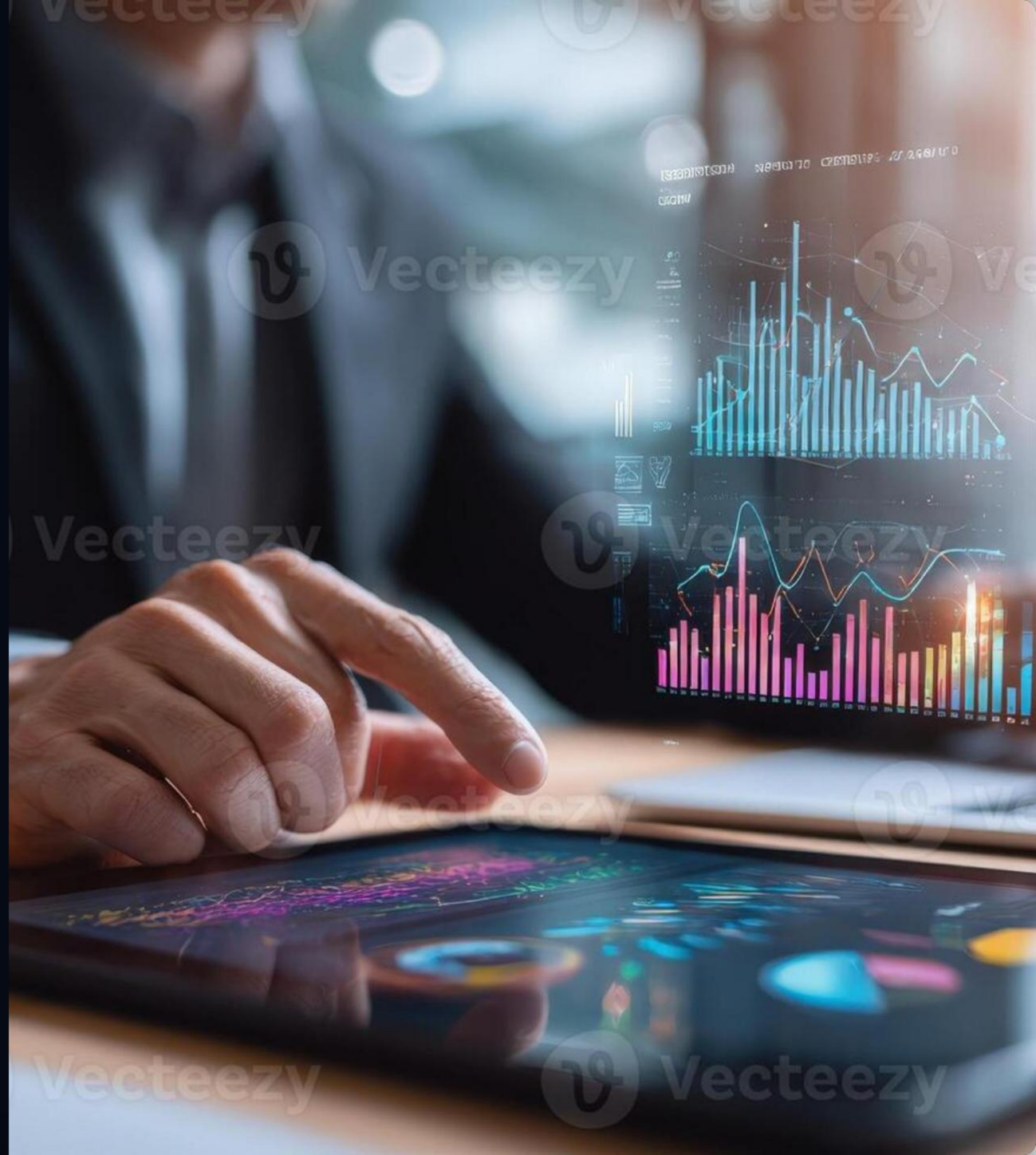
STRATEGIC ROADMAP

- **Immediate Tech Outreach:** Automate high-priority support for users with escalating Friction Scores.
- **Plan Right-Sizing:** Proactively transition users with high Charge Velocity to stable tiers.
- **Surprise & Delight:** Targeted loyalty rewards for "Silent Risk" customers to reinvigorate brand engagement.

FINANCIAL ROI

Transitioning to our proactive machine learning model allows ConnectTel to mitigate churn by an estimated 25% within the top value segment.

$2.4M USD
Estimated Annual Revenue Preserved



QUESTIONS?

ConnectTel Data Strategy Roadmap

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IMAGE SOURCES

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Source: www.vecteezy.com